

CS 532 HW5

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Problem

Consider three LANs interconnected by two routers, as shown in Figure 6.33.

1. Assign IP addresses to all of the interfaces. For Subnet 1 use addresses of the form 192.168.1.xxx; for Subnet 2 uses addresses of the form 192.168.2.xxx; and for Subnet 3 use addresses of the form 192.168.3.xxx.
2. Consider sending an IP datagram from Host E to Host B. Suppose all of the ARP tables are up to date. Enumerate all the steps, as done for the single-router example in Section 6.4.1.

Solution

1. We use /24 subnets with the required base prefixes.
 - (a) Subnet 1: 192.168.1.0/24
 - Host A: 192.168.1.1
 - Host B: 192.168.1.2
 - R1 interface on Subnet 1: 192.168.1.3
 - (b) Subnet 2: 192.168.2.0/24
 - Host C: 192.168.2.1
 - Host D: 192.168.2.2
 - R1 interface on Subnet 2: 192.168.2.3
 - R2 interface on Subnet 2: 192.168.2.4
 - (c) Subnet 3: 192.168.3.0/24
 - Host E: 192.168.3.1
 - Host F: 192.168.3.2
 - R2 interface on Subnet 3: 192.168.3.3

2. Let the following symbolic names denote MAC and IP addresses:

$$IP_E = 192.168.3.1, \quad MAC_E = E's \text{ adapter MAC}, \quad (1)$$

$$IP_B = 192.168.1.2, \quad MAC_B = B's \text{ adapter MAC}, \quad (2)$$

$$IP_{R2,3} = 192.168.3.3, \quad MAC_{R2,3} = R2's \text{ MAC on Subnet 3}, \quad (3)$$

$$IP_{R2,2} = 192.168.2.4, \quad MAC_{R2,2} = R2's \text{ MAC on Subnet 2}, \quad (4)$$

$$IP_{R1,2} = 192.168.2.3, \quad MAC_{R1,2} = R1's \text{ MAC on Subnet 2}, \quad (5)$$

$$IP_{R1,1} = 192.168.1.3, \quad MAC_{R1,1} = R1's \text{ MAC on Subnet 1}. \quad (6)$$

Step-by-step forwarding process:

- (a) **At host E:** E creates an IP datagram with source IP_E and destination IP_B . Since IP_B is outside Subnet 3, E forwards to its default gateway $IP_{R2,3}$.
- (b) **E sends an Ethernet frame on Subnet 3:** Using ARP, E already knows $MAC_{R2,3}$. The frame fields are:

$$\text{Src MAC} = MAC_E, \quad \text{Dst MAC} = MAC_{R2,3}.$$

Payload: the IP datagram (E→B).

- (c) **R2 receives the frame:** R2 accepts it, removes the Ethernet header, and examines the IP datagram. Destination 192.168.1.2 is in Subnet 1, reachable via R1.
- (d) **R2 forwards onto Subnet 2:** R2 uses next hop $IP_{R1,2}$. Ethernet frame on Subnet 2:

$$\text{Src MAC} = MAC_{R2,2}, \quad \text{Dst MAC} = MAC_{R1,2}.$$

- (e) **R1 receives the frame on Subnet 2:** R1 strips the Ethernet header and inspects the IP datagram. Destination is directly reachable on Subnet 1.
- (f) **R1 forwards onto Subnet 1:** ARP gives MAC_B . New Ethernet frame:

$$\text{Src MAC} = MAC_{R1,1}, \quad \text{Dst MAC} = MAC_B.$$

- (g) **Host B receives the frame:** B accepts the frame, removes the Ethernet header, and processes the IP datagram (destination IP matches IP_B).

The IP source and destination never change: always E→B.

The MAC addresses change on every hop:

$$E \rightarrow R2 \quad R2 \rightarrow R1 \quad R1 \rightarrow B.$$

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Problem

In this problem, you will put together much of what you have learned about Internet protocols. Suppose you walk into a room, connect to Ethernet, and want to download a Web page. What are all the protocol steps that take place, starting from powering on your PC to getting the Web page? Assume there is nothing in our DNS or browser caches when you power on your PC. (Hint: The steps include the use of Ethernet, DHCP, ARP, DNS, TCP, and HTTP protocols.) Explicitly indicate in your steps how you obtain the IP and MAC addresses of a gateway router.

Solution

We assume an empty ARP cache, empty DNS cache, and an Ethernet connection. The steps involve Ethernet, DHCP, ARP, DNS, TCP, and HTTP.

1. Ethernet Link Activation

- (a) The user connects the Ethernet cable.
- (b) The NIC and switch port perform Ethernet autonegotiation (speed/duplex).
- (c) The link becomes active. The host knows its own MAC address, but has no IP address yet.

2. Obtaining an IP Address and Default Gateway (DHCP)

- (a) The host sends a *DHCPDISCOVER* message:
 - Ethernet: broadcast destination MAC `ff:ff:ff:ff:ff:ff`.
 - IP: source `0.0.0.0`, destination `255.255.255.255`.
 - UDP ports: `68` → `67`.
- (b) The DHCP server replies with a *DHCPOFFER* containing:
 - Assigned IP address (e.g., `192.168.1.10`)
 - Subnet mask (e.g., `255.255.255.0`)
 - Default gateway IP (e.g., `192.168.1.1`)
 - DNS server IP address
- (c) The host accepts via *DHCPREQUEST* and receives *DHCPACK*.

At this point the host knows:

$$IP_{\text{host}} = 192.168.1.10, \quad GW_{\text{IP}} = 192.168.1.1.$$

4. Learning the Gateway's MAC Address (ARP)

- (a) The host must send any off-subnet traffic through the default gateway. To send an Ethernet frame to the gateway, it needs the gateway's MAC address.

- (b) The host broadcasts an ARP Request:
“Who has 192.168.1.1? Tell 192.168.1.10.”
- (c) The gateway responds with an ARP Reply:

$$192.168.1.1 \longrightarrow \text{MAC}_{\text{GW}}.$$

- (d) The host stores the mapping

$$(192.168.1.1 \mapsto \text{MAC}_{\text{GW}})$$

in its ARP table.

7. Resolving the Web Server’s Domain Name (DNS)

- (a) The browser receives a URL such as `http://www.example.com`. DNS and browser caches are empty.
- (b) The host constructs a DNS query asking for the A record of `www.example.com`. The DNS server IP was provided by DHCP.
- (c) To send the DNS query:
 - IP header: `src = 192.168.1.10`, `dst = DNS server IP`.
 - Ethernet header: `dst MAC = MACGW` (off-subnet).
- (d) The DNS server replies with the IP address of the web server, e.g.,

$$\text{IP}_{\text{www}} = 93.184.216.34.$$

11. Establishing a TCP Connection to the Web Server

- (a) The browser initiates a TCP three-way handshake:
 - Host sends TCP SYN:

$$\text{Src IP} = 192.168.1.10, \quad \text{Dst IP} = 93.184.216.34.$$
 - Ethernet: `dst MAC = MACGW` (web server is off-subnet).
- (b) Routers forward the SYN across the Internet. The web server responds with SYN-ACK.
- (c) The host responds with ACK. TCP connection is now established.

15. HTTP Request and Page Download

- (a) The browser sends an HTTP GET request over the TCP connection:

```
GET / HTTP/1.1
Host: www.example.com
```

- (b) The web server sends the HTTP response:
 - Status line (e.g., `HTTP/1.1 200 OK`)

- Response headers
 - HTML content
- (c) Additional HTTP requests (images, CSS, JavaScript) are made as needed. All traffic continues using the same IP-to-MAC mapping for the default gateway.

Summary of Protocol Roles:

- **Ethernet:** Local delivery of all frames (DHCP, ARP, DNS, TCP/IP, HTTP).
- **DHCP:** Obtains host IP, subnet mask, gateway IP, and DNS server IP.
- **ARP:** Obtains gateway MAC address.
- **DNS:** Resolves the web host's domain name to an IP address.
- **TCP:** Establishes a reliable connection to the web server.
- **HTTP:** Requests and receives the actual web page.